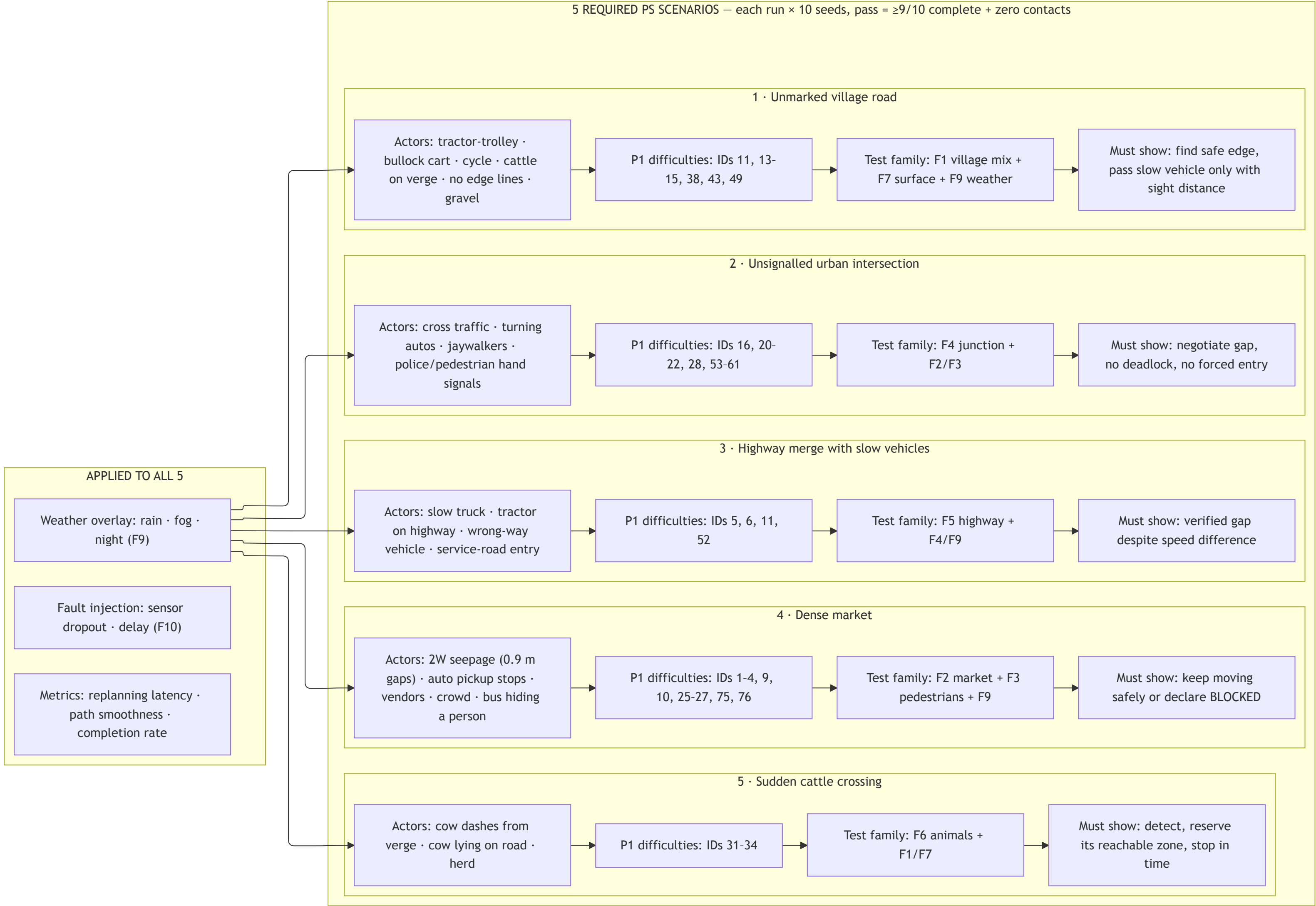


D4 · The 5 required Indian scenarios

Who is on the road → which Indian difficulties (P1 IDs) → test family → what the car must show.



D4 · The 5 required Indian scenarios — exact spec for redrawing

Redraw rule for Gemini / Nano Banana: keep every box, group and arrow listed. Do not add, merge or drop any. Styling is free; structure is fixed.

BOXES (grouped exactly as drawn):

- 5 REQUIRED PS SCENARIOS — each run × 10 seeds, pass = $\geq 9/10$ complete + zero contacts
 - 1 · Unmarked village road
 - [Va] Actors: tractor-trolley · bullock cart · cycle · cattle on verge · no edge lines · gravel
 - [Vi] P1 difficulties: IDs 11, 13–15, 38, 43, 49
 - [Vf] Test family: F1 village mix + F7 surface + F9 weather
 - [Vp] Must show: find safe edge, pass slow vehicle only with sight distance
 - 2 · Unsignalled urban intersection
 - [Ua] Actors: cross traffic · turning autos · jaywalkers · police/pedestrian hand signals
 - [Ui] P1 difficulties: IDs 16, 20–22, 28, 53–61
 - [Uf] Test family: F4 junction + F2/F3
 - [Up] Must show: negotiate gap, no deadlock, no forced entry
 - 3 · Highway merge with slow vehicles
 - [Ha] Actors: slow truck · tractor on highway · wrong-way vehicle · service-road entry
 - [Hi] P1 difficulties: IDs 5, 6, 11, 52
 - [Hf] Test family: F5 highway + F4/F9
 - [Hp] Must show: verified gap despite speed difference
 - 4 · Dense market
 - [Ma] Actors: 2W seepage (0.9 m gaps) · auto pickup stops · vendors · crowd · bus hiding a person
 - [Mi] P1 difficulties: IDs 1–4, 9, 10, 25–27, 75, 76
 - [Mf] Test family: F2 market + F3 pedestrians + F9
 - [Mp] Must show: keep moving safely or declare BLOCKED
 - 5 · Sudden cattle crossing
 - [Ca] Actors: cow dashes from verge · cow lying on road · herd
 - [Ci] P1 difficulties: IDs 31–34
 - [Cf] Test family: F6 animals + F1/F7
 - [Cp] Must show: detect, reserve its reachable zone, stop in time
- APPLIED TO ALL 5
 - [ov1] Weather overlay: rain · fog · night (F9)
 - [ov2] Fault injection: sensor dropout · delay (F10)
 - [ov3] Metrics: replanning latency · path smoothness · completion rate

CONNECTIONS (20 arrows, from → to : what flows):

Actors: tractor-trolley · bullock cart · cycle · cattle on verge · no edge lines · gravel → P1
difficulties: IDs 11, 13–15, 38, 43, 49
P1 difficulties: IDs 11, 13–15, 38, 43, 49 → Test family: F1 village mix + F7 surface + F9 weather
Test family: F1 village mix + F7 surface + F9 weather → Must show: find safe edge, pass slow vehicle only with sight distance
Actors: cross traffic · turning autos · jaywalkers · police/pedestrian hand signals → P1
difficulties: IDs 16, 20–22, 28, 53–61
P1 difficulties: IDs 16, 20–22, 28, 53–61 → Test family: F4 junction + F2/F3
Test family: F4 junction + F2/F3 → Must show: negotiate gap, no deadlock, no forced entry
Actors: slow truck · tractor on highway · wrong-way vehicle · service-road entry → P1 difficulties: IDs 5, 6, 11, 52
P1 difficulties: IDs 5, 6, 11, 52 → Test family: F5 highway + F4/F9
Test family: F5 highway + F4/F9 → Must show: verified gap despite speed difference
Actors: 2W seepage (0.9 m gaps) · auto pickup stops · vendors · crowd · bus hiding a person → P1
difficulties: IDs 1–4, 9, 10, 25–27, 75, 76
P1 difficulties: IDs 1–4, 9, 10, 25–27, 75, 76 → Test family: F2 market + F3 pedestrians + F9
Test family: F2 market + F3 pedestrians + F9 → Must show: keep moving safely or declare BLOCKED
Actors: cow dashes from verge · cow lying on road · herd → P1 difficulties: IDs 31–34
P1 difficulties: IDs 31–34 → Test family: F6 animals + F1/F7
Test family: F6 animals + F1/F7 → Must show: detect, reserve its reachable zone, stop in time
Weather overlay: rain · fog · night (F9) → Actors: tractor-trolley · bullock cart · cycle · cattle on verge · no edge lines · gravel
Weather overlay: rain · fog · night (F9) → Actors: cross traffic · turning autos · jaywalkers · police/pedestrian hand signals
Weather overlay: rain · fog · night (F9) → Actors: slow truck · tractor on highway · wrong-way vehicle · service-road entry
Weather overlay: rain · fog · night (F9) → Actors: 2W seepage (0.9 m gaps) · auto pickup stops · vendors · crowd · bus hiding a person
Weather overlay: rain · fog · night (F9) → Actors: cow dashes from verge · cow lying on road · herd

Mermaid source (paste into mermaid.live to re-render)

```
graph LR
    subgraph PS ["5 REQUIRED PS SCENARIOS — each run × 10 seeds, pass = ≥9/10 complete + zero contacts"]
        subgraph Vg["1 · Unmarked village road"]
            Va["Actors: tractor-trolley · bullock cart · cycle · cattle on verge · no edge lines · gravel"]
            Vi["P1 difficulties: IDs 11, 13–15, 38, 43, 49"]
        end
    end
```

```

Vf["Test family: F1 village mix + F7 surface + F9 weather"]
Vp["Must show: find safe edge, pass slow vehicle only with sight distance"]
end
subgraph Ug["2 · Unsignalled urban intersection"]
  Ua["Actors: cross traffic · turning autos · jaywalkers · police/pedestrian hand signals"]
  Ui["P1 difficulties: IDs 16, 20-22, 28, 53-61"]
  Uf["Test family: F4 junction + F2/F3"]
  Up["Must show: negotiate gap, no deadlock, no forced entry"]
end
subgraph Hg["3 · Highway merge with slow vehicles"]
  Ha["Actors: slow truck · tractor on highway · wrong-way vehicle · service-road entry"]
  Hi["P1 difficulties: IDs 5, 6, 11, 52"]
  Hf["Test family: F5 highway + F4/F9"]
  Hp["Must show: verified gap despite speed difference"]
end
subgraph Mg["4 · Dense market"]
  Ma["Actors: 2W seepage (0.9 m gaps) · auto pickup stops · vendors · crowd · bus hiding a person"]
  Mi["P1 difficulties: IDs 1-4, 9, 10, 25-27, 75, 76"]
  Mf["Test family: F2 market + F3 pedestrians + F9"]
  Mp["Must show: keep moving safely or declare BLOCKED"]
end
subgraph Cg["5 · Sudden cattle crossing"]
  Ca["Actors: cow dashes from verge · cow lying on road · herd"]
  Ci["P1 difficulties: IDs 31-34"]
  Cf["Test family: F6 animals + F1/F7"]
  Cp["Must show: detect, reserve its reachable zone, stop in time"]
end
end
subgraph OV["APPLIED TO ALL 5"]
  ov1["Weather overlay: rain · fog · night (F9)"]
  ov2["Fault injection: sensor dropout · delay (F10)"]
  ov3["Metrics: replanning latency · path smoothness · completion rate"]
end
Va --> Vi
Vi --> Vf
Vf --> Vp
Ua --> Ui
Ui --> Uf
Uf --> Up
Ha --> Hi
Hi --> Hf
Hf --> Hp
Ma --> Mi
Mi --> Mf
Mf --> Mp
Ca --> Ci
Ci --> Cf
Cf --> Cp
ov1 --> Va
ov1 --> Ua
ov1 --> Ha
ov1 --> Ma
ov1 --> Ca

```